XINJE PLC — REVISED I/O LIST

Two-Station Conveyor Sorting System | Includes barcode classification via Modbus TCP

Revision: Barcode signals have been moved out of the physical input list and into a separate Modbus communication mapping. This is a proposed logical design based on your OpenPLC variable screenshots; the exact XINJE model and Modbus register addresses are not yet confirmed.

1. Physical Digital Inputs (X)

Proposed address	Program variable	Physical device	Purpose
X0	Start_PB	Start push button	Start request
X1	Stop_PB	Stop push button	Stop request
X2*	Emergency_OK	Safety circuit status contact	Healthy/permissive status only
X3	PE_Entrance	Entrance photoelectric sensor	Detect incoming package
X4	St1_Left_PE	Station 1 left photoelectric sensor	Package position / detection
X5	St1_Right_PE	Station 1 right photoelectric sensor	Package position / detection
X6	St1_Left_ResetPB	Station 1 left reset push button	Reset left counter
X7	St1_Right_ResetPB	Station 1 right reset push button	Reset right counter
X10	St2_Left_PE	Station 2 left photoelectric sensor	Package position / detection
X11	St2_Right_PE	Station 2 right photoelectric sensor	Package position / detection
X12	St2_Left_ResetPB	Station 2 left reset push button	Reset left counter
X13	St2_Right_ResetPB	Station 2 right reset push button	Reset right counter
X14	VFD_Fault_OK	VFD healthy/fault relay contact	Drive status feedback

*Emergency stop must be implemented through an appropriate hardwired safety circuit/safety relay. A standard PLC input alone is not a safety function.

2. Physical Digital Outputs (Y)

Proposed address	Program variable	Connected device	Purpose
Y0	Do_VFD_Start	VFD digital input	Conveyor start command
Y1	Sol_St1_Left	Station 1 left solenoid valve	Divert package left
Y2	Sol_Str1_Right	Station 1 right solenoid valve	Divert package right
Y3	Sol_Str2_Left	Station 2 left solenoid valve	Divert package left
Y4	Sol_Str2_Right	Station 2 right solenoid valve	Divert package right
Y5	St1_Left_Lamp	Station 1 left lamp	Left operation indication
Y6	St1_Right_Lamp	Station 1 right lamp	Right operation indication
Y7	St2_Left_Lamp	Station 2 left lamp	Left operation indication
Y10	St2_Right_Lamp	Station 2 right lamp	Right operation indication
Y11*	Alarm_VFD_Tripped	Alarm lamp/buzzer	Optional physical alarm output
Y12*	Camera_Rec	Camera/interface trigger	Optional, only if physically required

*Optional outputs. If Conveyor_Run is only a software status bit or duplicates Do_VFD_Start, do not allocate a separate output unless you need a separate physical signal.

3. Barcode Classification — Modbus TCP (Logical Signals)

These four signals are received through Modbus TCP. They are not physical X inputs when they are mapped from communication data. The PLC program should read the configured Modbus coils/registers and copy or map the received values into the named BOOL variables.

OpenPLC variable	Meaning	PLC program mapping
St1_Sig_Left	Station 1 left route selected	Modbus data → internal BOOL
St1_Sig_Right	Station 1 right route selected	Modbus data → internal BOOL
St2_Sig_Left	Station 2 left route selected	Modbus data → internal BOOL
St2_Sig_Right	Station 2 right route selected	Modbus data → internal BOOL

Modbus address note: Exact coil/register numbers cannot be assigned reliably until the communication roles and XINJE model are known. Confirm whether the XINJE PLC is the Modbus TCP client/master or server/slave, and which device publishes the barcode classification bits. Avoid assuming that a Modbus address is the same as a physical X address.

4. Internal Variables — No Physical Terminal Required

Variable	Type	Purpose
System_Run	BOOL	System run latch/status
St1_Busy / St2_Busy	BOOL	Station occupancy status
St1_Flag_Left / St1_Flag_Right	BOOL	Station 1 sequence flags
St2_Flag_Left / St2_Flag_Right	BOOL	Station 2 sequence flags
St1_Left_Full / St1_Right_Full	BOOL	Station 1 bin-full flags
St2_Left_Full / St2_Right_Full	BOOL	Station 2 bin-full flags
St1_Left_Count / St1_Right_Count	INT	Station 1 package counts
St2_Left_Count / St2_Right_Count	INT	Station 2 package counts
St1_Left_CTU / St1_Right_CTU	CTU	Station 1 counter instances
St2_Left_CTU / St2_Right_CTU	CTU	Station 2 counter instances
St1_Timer_Left/Right; St2_Timer_Left/Right	TP	Timer instances
St1_Busy_Timer / St2_Busy_Timer	TP	Busy timer instances
FB_VFD1	FB_VFD	VFD function-block instance
A0	INT	Purpose unclear from screenshot; confirm whether intended for analog data

5. Hardware and Wiring Checks

• Confirm the exact XINJE PLC model and physical terminal layout before assigning real terminal numbers.

• Verify input common wiring, sensor supply voltage, and PNP/NPN compatibility.

• Check whether the PLC outputs are relay or transistor type and confirm voltage/current ratings. Use suitable interposing relays/drivers for solenoid loads where required.

• Configure the VFD digital input and common/reference correctly; use a separate status contact for VFD fault feedback.

• Barcode scanner data may require an interface such as an ESP32 depending on scanner and PLC protocol compatibility. Modbus TCP carries mapped data, not a direct physical X input.

• The screenshots show some additional New_Scan_Trigger bits; treat them as internal or communication variables unless you confirm they are physically wired signals.

Status: Revised proposed worksheet. Confirm model, protocol roles, and register mapping before wiring or commissioning.